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EXTENDED ABSTRACT

Statistical downscaling for precipitation examinating potential temporal dependence due to climate change

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Keywords: Statistical climate downscaling, Bernoulli-Gamma Model, Temporal variation, Non-stationarity, Kernel-based weighted likelihood

1. Introduction and objectives

The Perfect Prognosis (PP) framework is one of the most widely used methods for statistical climate downscaling, in which the empirical relationships between large-scale variables (e.g., from reanalysis data) and local observations are learned (Klein et al., 1959; Vaittinada Ayar et al., 2016). Within this framework, the generalized linear models (GLMs) based on the Bernoulli-Gamma distribution have been shown to be efficient in modelling precipitation (Stern and Coe, 1984; Chandler and Wheater, 2002; Vrac et al., 2007). However, most existing PP-based downscaling models typically assume time invariance, which might be inadequate over a large time scale in the context of climate change (Maraun et al., 2010).

In this work, we propose a dynamic modeling strategy to analyze the potential temporal variation in the downscaling parameters of the commonly used Bernoulli-Gamma GLM. Specifically, we adopt a kernel-based weighted likelihood approach that allows model parameters to evolve smoothly over time (Cai et al., 2000). The objective is to assess whether the downscaling parameters present significant temporal variability or long-term trends, rather than remaining time-invariant. Identifying such non-stationary behaviors could provide insights into the development of more flexible statistical downscaling models in a changing climate scenario by relaxing the time stationary assumption.

2. Methods and data

The proposed approach is based on a Bernoulli-Gamma GLM for precipitation downscaling (as shown in Fig. 1), which models both precipitation occurrence (with a Bernoulli distribution) and intensity (with a Gamma distribution). To account for potential temporal non-stationarity, model parameters are estimated using a kernel-based weighted likelihood approach, in which observations are weighted according to their temporal proximity to a target date. Several kernel functions are considered, including truncated Gaussian kernels (shown in Fig. 1), allowing flexible control of temporal smoothing.

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The new model is evaluated using open-source climate datasets. Large-scale atmospheric predictors such as dew point temperature and wind speed are obtained from the ERA5 reanalysis data (Hersbach et al., 2020), while local-scale precipitation observations are drawn from the E-OBS gridded dataset[1].

3. Preliminary results and/or expected outcomes

As illustrated in Fig. 1, the inputs to our model consist of large-scale predictors and local-scale precipitation, covering the period from 1950 to present. The Bernoulli-Gamma GLM is employed for downscaling, incorporating multiple covariates, each influencing the distribution parameters via $\{a_j, b_j\}, j = 1, \ldots, p$. By applying a kernel-based weight with a moving time window and varying bandwidths on the current model, we aim to examine whether accounting for temporal variation in the parameters can effectively reveal the non-stationary climate behaviors.

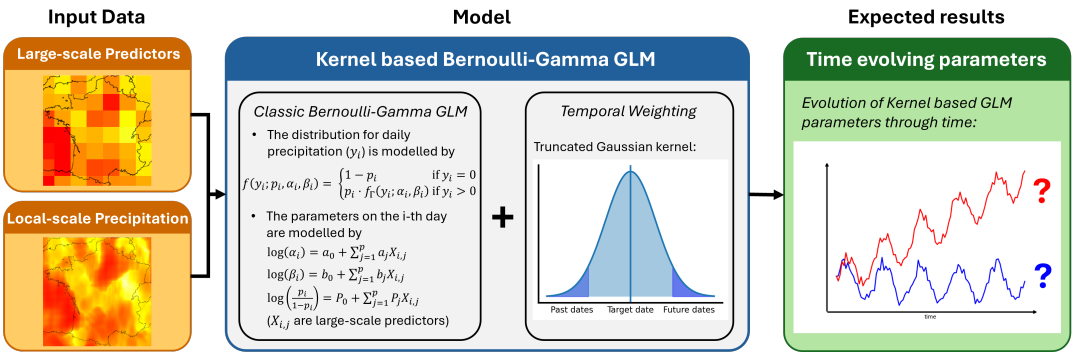

Figure 1. *Visualization of the research framework.*

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[1] https://www.ecad.eu/download/ensembles/ensembles.php